HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics and Statistics

F70LB

F70LB LIFE INSURANCE MATHEMATICS B

Semester 2 2020/21

Duration: Two Hours

Candidates must attempt ALL questions.
This is an open book exam but all answers must be in your own words to demonstrate your understanding. You must also show your working.
Answers that do not demonstrate understanding or show working may not receive marks.
Formulae and Tables for Examinations 2002 may be used.
You may use a permitted calculator.

1.

(a) A life insurer is planning to launch a new five year endowment assurance. There is some debate over the final product design so the following results have been produced for two options, A and B.

Option	Profit signature
A	(-1,125, 345, 505, 531, 300)
B	(-603, 146, 256, 224, 451)

The insurer wishes to proceed with the product with the shortest discounted payback period, using a risk discount rate of 9% per annum. Determine which design the insurer should choose.

(5 marks)

(b) Explain why the discounted payback period is a useful profit criterion for the insurer.

(1 mark)

(c) The design that you have identified in (a) will be sold to policyholders aged 35, for a premium of £2,000 payable annually in advance. Calculate the profit margin that the insurer will receive on the contract, on the following basis:

Mortality	AM92 select
Risk discount rate	9% p.a.

(5 marks)

[Total 11 marks]

2. An insurer sells Long-Term Care (LTC) policies. It uses a 4-state Markov model to price these policies in which state 1 is the healthy state and state 4 is the dead state.

(a) Draw a 4-state Markov model that could be used by an insurer for these policies with appropriate labels for each state and necessary transitional intensity arrows resulting ${}_tp_x^{11} = {}_tp_x^{\overline{11}}$, but ${}_tp_x^{33} \neq {}_tp_x^{\overline{33}}$.

(2 marks)

(b) For $j = 2$ and $j = 4$, state with brief reasons whether ${}_tp_x^{jj} = {}_tp_x^{\overline{jj}}$ or ${}_tp_x^{jj} \neq {}_tp_x^{\overline{jj}}$.

(2 marks)

(c) State an expression for ${}_tp_x^{31}$ and, without doing any calculations, briefly explain why your expression holds.

(1 mark)

(d) Suggest, with brief reasons, in what circumstances it might be reasonable for ${}_tp_x^{11} = {}_tp_x^{\overline{11}}$

(1 mark)

[Total 6 marks]

3. A woman aged 59 and her 60 year old husband wish to buy a policy which provides the following benefits:

   - a lifetime annuity starting at £30,000 per annum payable annually in advance while they are both alive, reducing to £20,000 once one of them dies; and
   - an annuity of £2,500 per annum payable annually in advance to each of their twin granddaughters, currently aged 11, commencing on the death of either grandparent and ceasing when the granddaughters reach age 21.

   (a) Assuming the following basis:

Mortality	Man - PMA92C20
	Wife - PFA92C20
	Granddaughters - Nil
Interest	4% per annum
Expenses	1% of the premium

      i. Calculate the granddaughters' benefit as the value of an annuity certain minus a joint-life temporary annuity, or otherwise.

      (6 marks)

      ii. Calculate the couple's benefit, then use the granddaughters' benefit from part (i) to calculate the single premium to be paid for the policy.

      (8 marks)

   (b) Comment on the appropriateness of the decision to ignore the granddaughters' mortality.

   (2 marks)

   [Total 16 marks]

4. Your department is developing a new unit-linked savings policy, which will not include any guaranteed death benefits.

   (a) A marketing colleague has suggested using only a fund management charge to recoup expenses, to make the policy easy to explain. Discuss this suggestion.

   (3 marks)

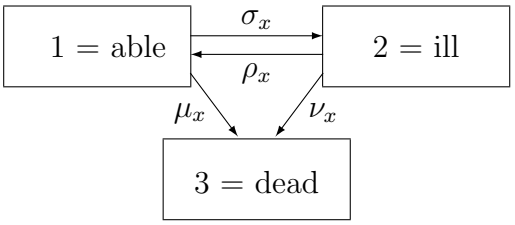
   (b) Suggest an alternative structure of charges that may be suitable for this contract, and give a reason for the inclusion of each charge suggested.

   (4 marks)

   (c) Briefly explain how a unitised with-profits policy differs from a unit-linked savings policy. **(2 marks)**

   [Total 9 marks]

5. A life insurer uses the following model for pricing its income protection policies:



(a) Write down the equation of value for the following contract sold to a healthy life age x, defining all terms used:

- Benefits of £25,000 per annum are payable continuously while the policyholder is unable to work due to sickness, reducing to £15,000 per annum after one year of benefit payments.
- A maximum benefit payment term of three years applies for any one period of sickness.
- A deferred period of three months applies for all new claims.
- Premiums at rate $\bar{P}$ per annum are payable continously up to age 65 years while healthy, and while sick with duration less than the deferred period.
- Premiums are waived while benefits are payable.
- Interest of $i\%$ per annum effective.
- Expenses are 3% of all premiums payable.

(6 marks)

(b) The above contract also includes a one month off-period. Explain what this means and why it is included in the policy design.

(2 marks)

[Total 8 marks]

6. Amanda has been a member of her employer's defined benefit pension scheme for 15 years, since she was aged 40. The scheme provides a retirement pension of 1/60th of Final Pensionable Salary for each year of service, where Final Pensionable Salary is the average salary earned over the three years before retirement.

   (a) Given Amanda's salaries have been £32,000, £36,000 and £41,000 in the last three years respectively and assuming there is no allowance for future salary increases, calculate the annual pension Amanda will receive if she stops working now and collects her pension at Normal Retirement Age.

   (2 marks)

   (b) The scheme guarantees to pay pension benefits for at least five years. On the basis given below, calculate the cost to the pension scheme of providing this guarantee for Amanda, assuming pensions are paid annually in advance and the Normal Retirement Age (NRA) is 65.

Mortality	PFA92C20
Interest	4% per annum

(3 marks)

[**Total 5 marks**]

END OF PAPER (TOTAL 55 marks)